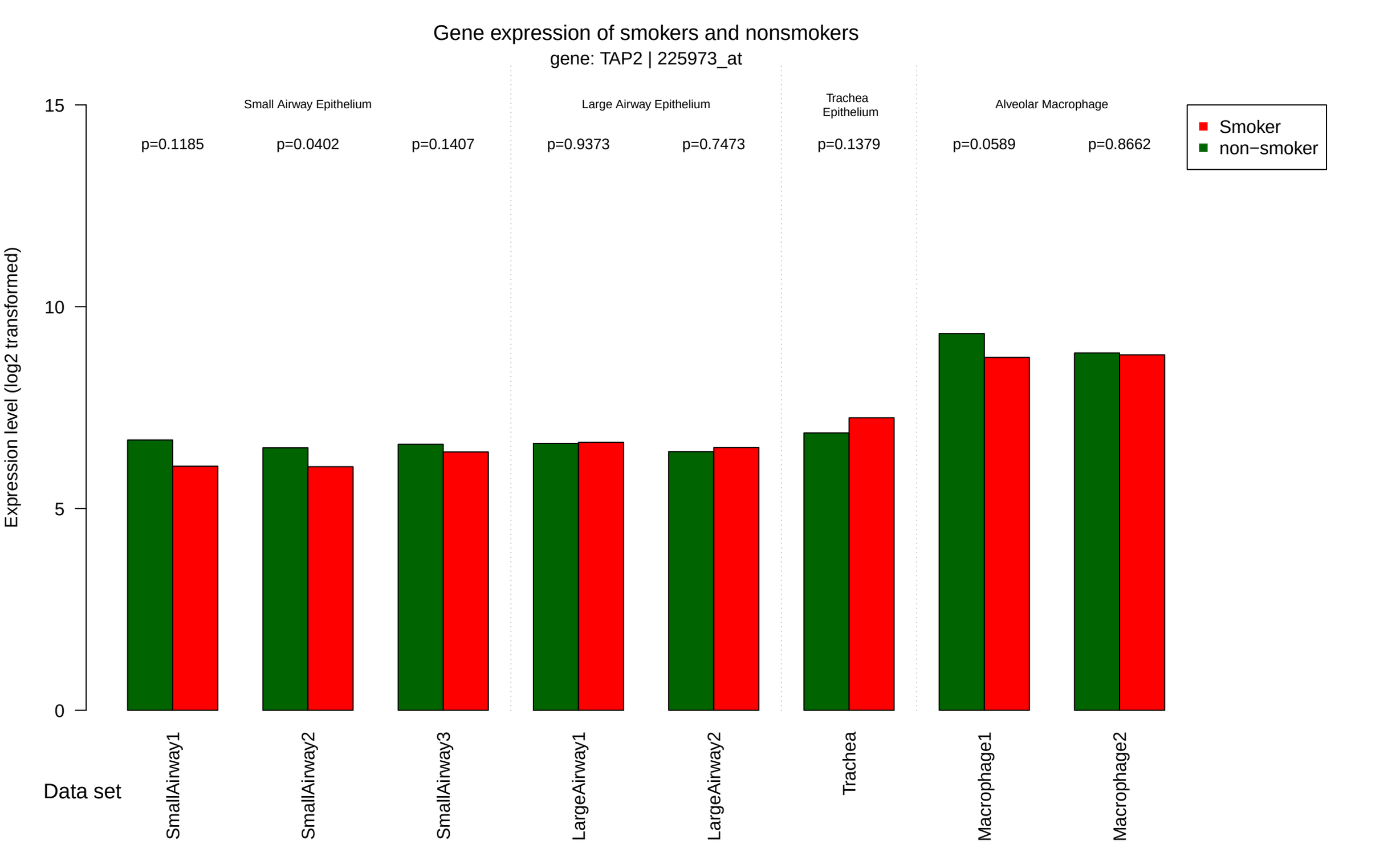
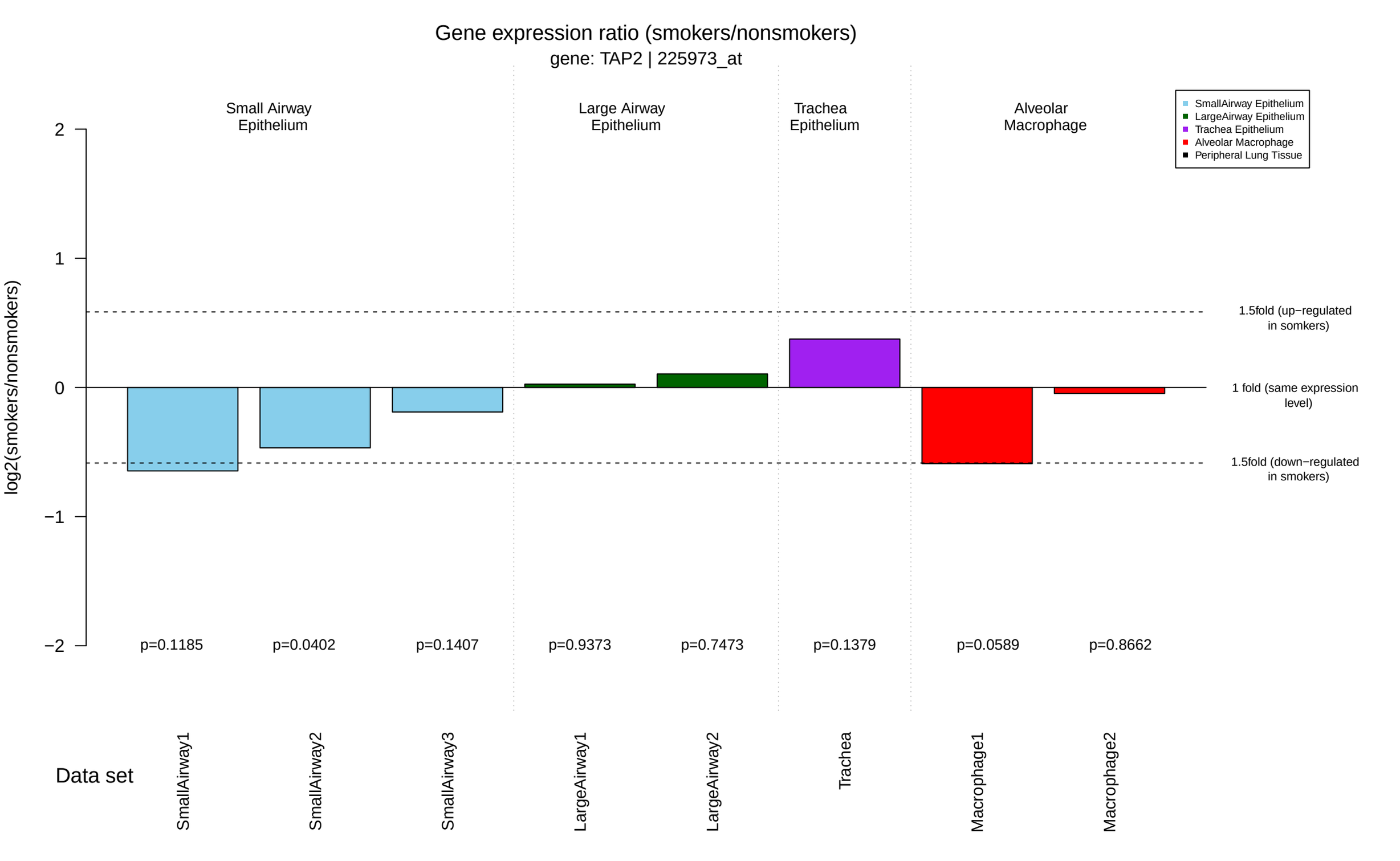
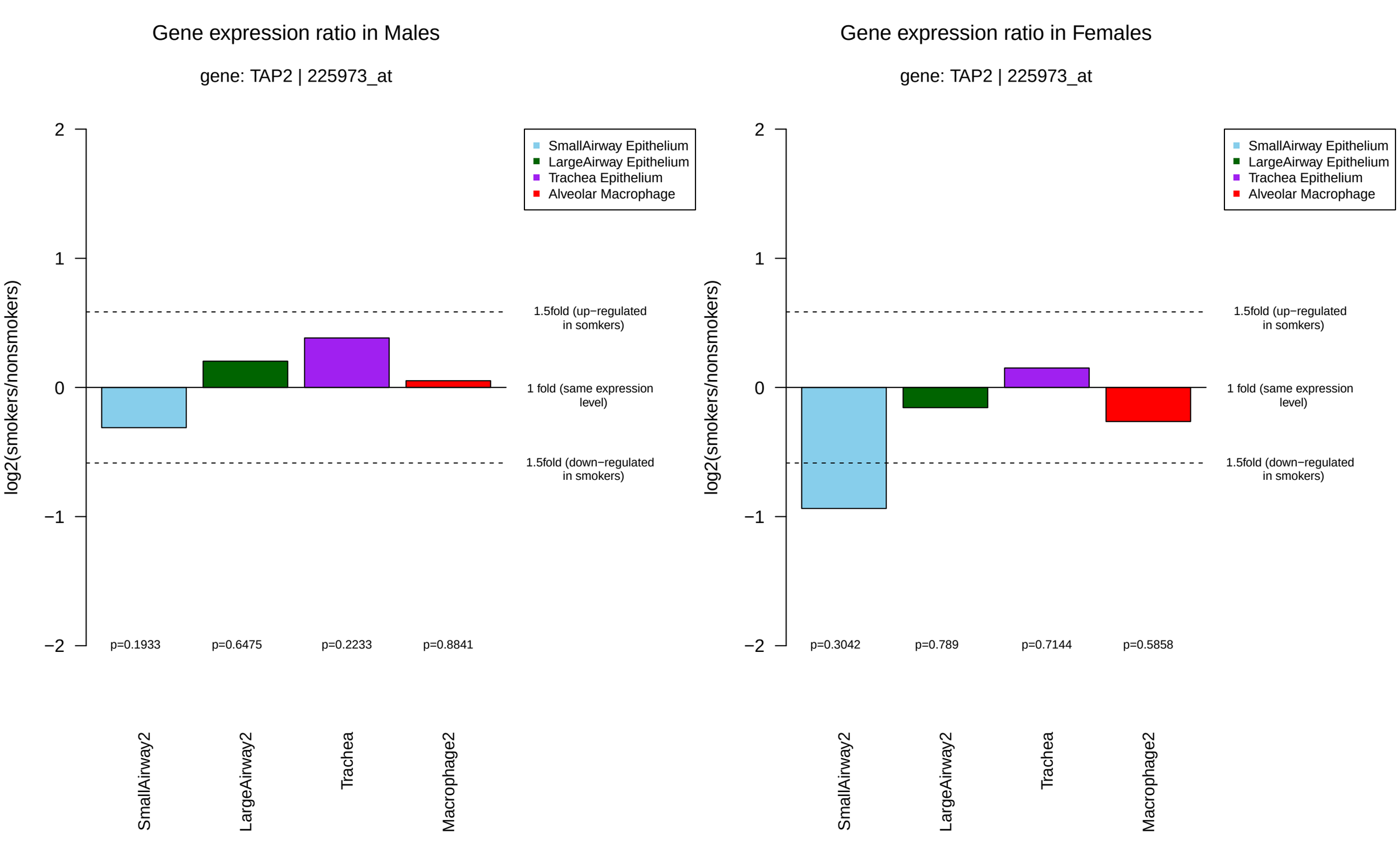
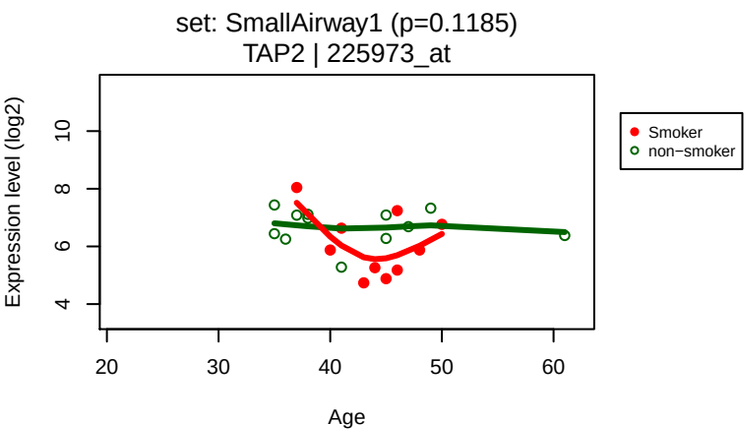
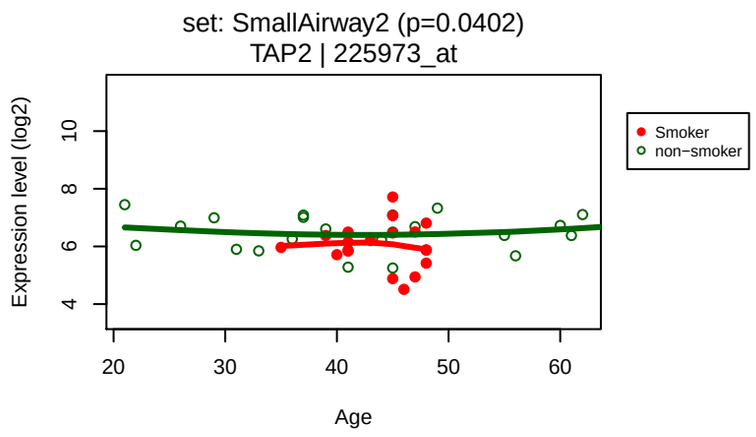
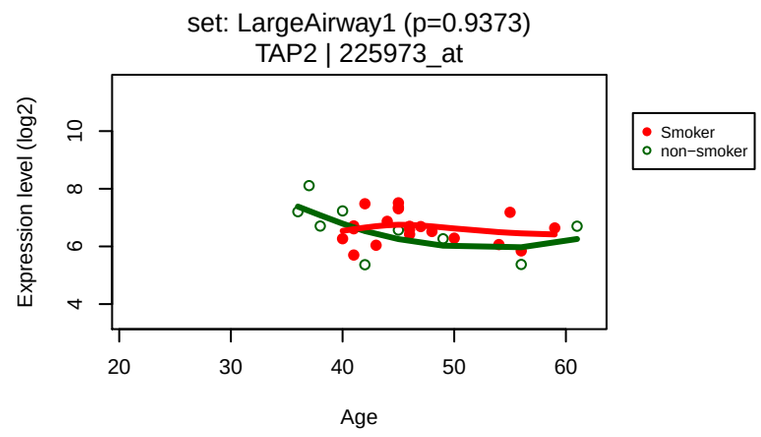

Gene expression of smokers and nonsmokers
gene: TAP2 | 225973_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1185
p=0.0402
p=0.1407
p=0.9373
p=0.7473
p=0.1379
p=0.0589
p=0.8662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

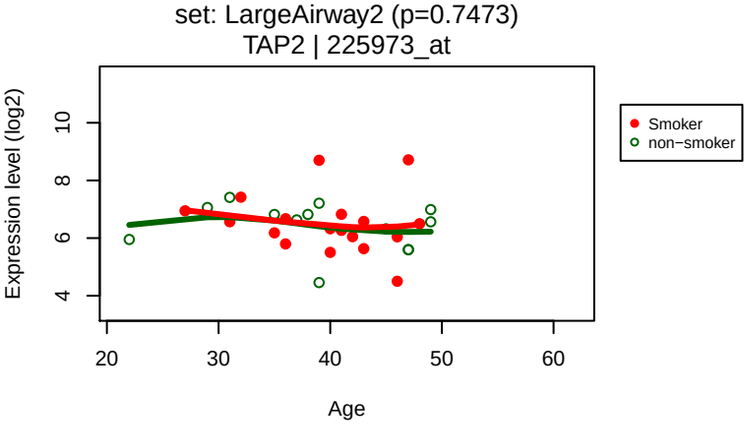

Gene expression ratio (smokers/nonsmokers)
gene: TAP2 | 225973_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1185
p=0.0402
p=0.1407
p=0.9373
p=0.7473
p=0.1379
p=0.0589
p=0.8662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

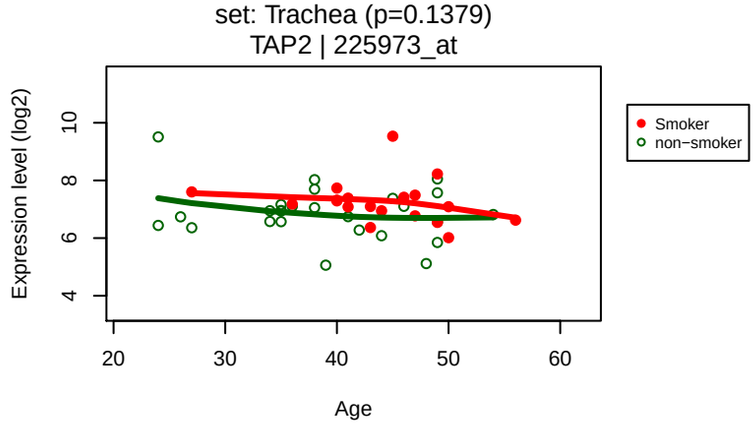

Gene expression ratio in Males
gene: TAP2 | 225973_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1933
p=0.6475
p=0.2233
p=0.8841
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAP2 | 225973_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3042
p=0.789
p=0.7144
p=0.5858
SmallAirway2
LargeAirway2
Trachea
Macrophage2

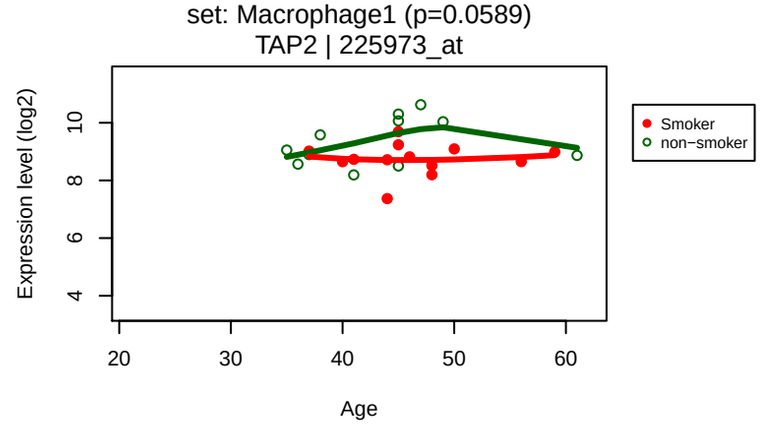

set: SmallAirway1 (p=0.1185)
TAP2 | 225973_at
Expression level (log2)
Age
Smoker
non−smoker

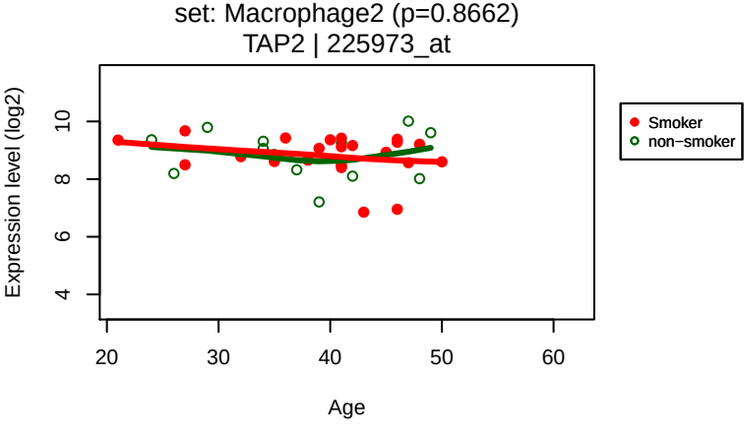

set: SmallAirway2 (p=0.0402)
TAP2 | 225973_at
Expression level (log2)
Age
Smoker
non−smoker

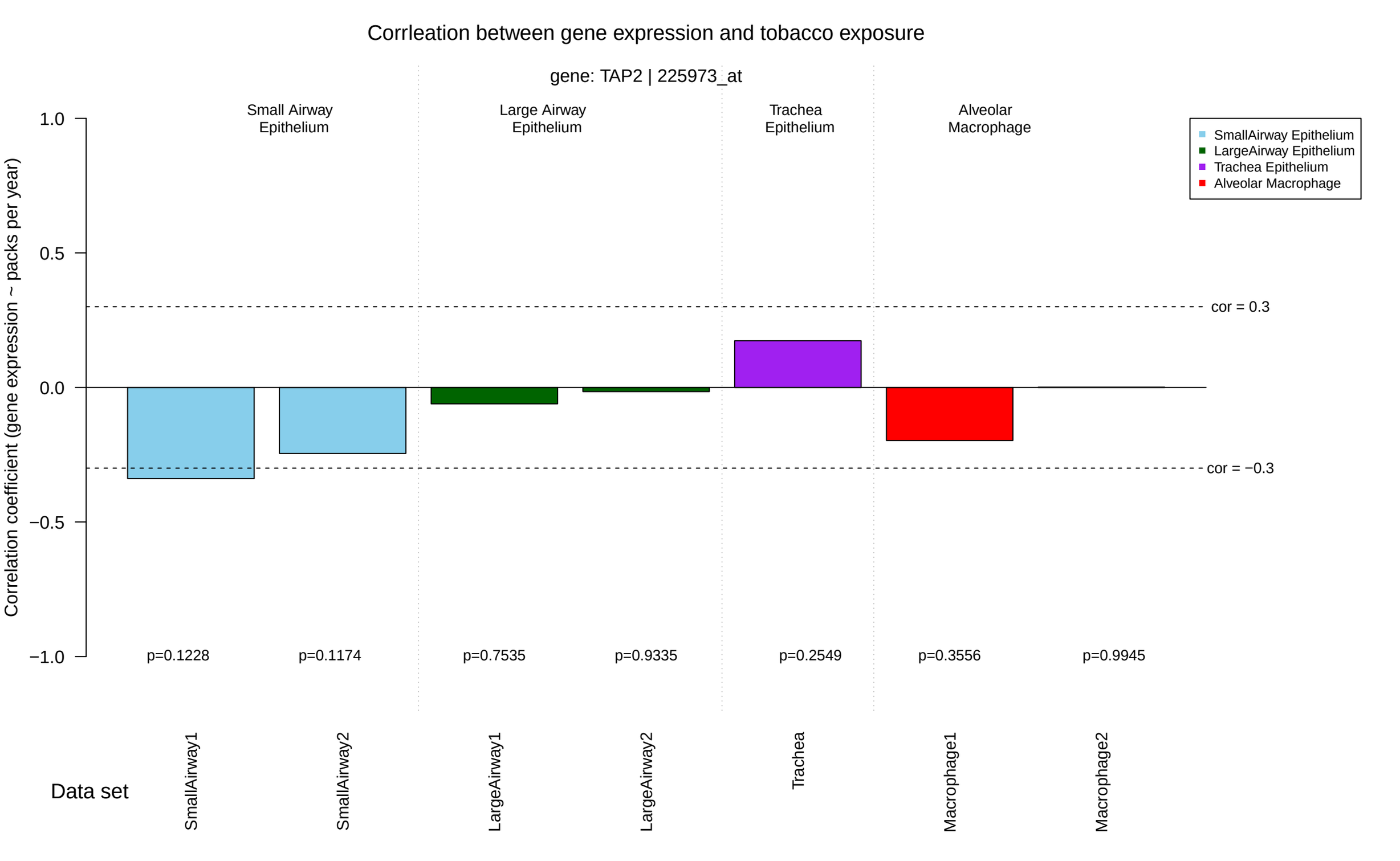

set: LargeAirway1 (p=0.9373)
TAP2 | 225973_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7473)
TAP2 | 225973_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1379)
TAP2 | 225973_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0589)
TAP2 | 225973_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8662)
TAP2 | 225973_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAP2 | 225973_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1228
p=0.1174
p=0.7535
p=0.9335
p=0.2549
p=0.3556
p=0.9945
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2